

Model-A Propagation-Aware Bragg Transfer Function

Square and Gaussian pulses with a worked 150 m case

MIGA programme

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Abstract

This manual documents the area-compensated propagation model implemented in the propagation-aware Python calculator. In a single-frequency retroreflected geometry, the forward optical envelope and its reflected copy reach the atoms with the round-trip separation $\tau_d = 2L/c$. Their overlap determines the effective two-photon Rabi envelope. Model A rescales every overlapped pulse to restore the nominal $\pi/2$ – π – $\pi/2$ areas, representing an experiment in which pulse areas are recalibrated after propagation is included. The manual derives the result for square and Gaussian optical-intensity pulses, explains the sensitivity and transfer functions, documents the numerical implementation, and gives a reproducible first-order example for $L = 150$ m, $T = 250$ ms, $\tau_\pi = 50$ μ s and $\tau_{\pi/2} = 25$ μ s. For the Gaussian case, these two widths are intensity FWHM values.

i Main result

Area compensation affects square and Gaussian pulses differently. A delayed square pulse becomes shorter and must become higher, which changes the high-frequency envelope of H_{AI} . An untruncated Gaussian remains a Gaussian of the same width; Model-A normalization exactly cancels its overlap attenuation, leaving only a shift by $\tau_d/2$. Consequently, the magnitude $|H_{AI}|$ is essentially unchanged for the Gaussian case, apart from negligible effects of numerical tail truncation.

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1 Purpose and model boundary

The calculator describes a resonant three-pulse Mach–Zehnder sequence with nominal areas $\pi/2-\pi-\pi/2$. It extends the finite-pulse sensitivity-function treatment of Cheinet *et al.*[1], the MIGA phase-noise analysis[2], and the pulse-shaping description based on accumulated Rabi area[3]. This manual considers only the area-compensated setting labelled A: area compensated in the interface.

The backend is an effective resonant two-level model. The Bragg order n multiplies the phase response but does not alter the normalized pulse dynamics. The calculation includes delayed envelope overlap, pulse-area recalibration, finite-pulse filtering and laser phase/frequency-noise conversion. It does not include the momentum-state ladder, velocity selectivity, diffraction phase, AC Stark shifts, Doppler detuning, spontaneous emission, parasitic interferometers or cavity dynamics. These effects require a multi-state Bragg treatment such as Refs. [4, 5].

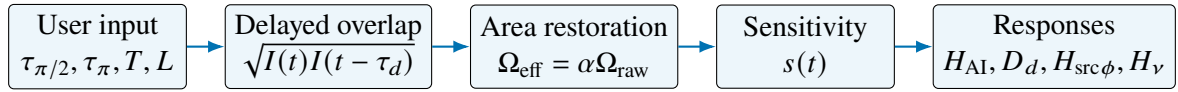


Figure 1. Model-A calculation chain. The delayed optical overlap determines a raw Rabi envelope. Its area is restored before the sensitivity and transfer functions are evaluated.

2 Quick start

Launch the interface from the project directory with

```
python3 bragg_transfer_function_propagation_ui.py
```

Select A: area compensated in the propagation-model menu. Choose Square or Gaussian, enter the $\pi/2$ and π widths in microseconds, and enter the centre-to-centre pulse separation T and one-way atom-to-retroreflector distance L . A square width is a command duration. A Gaussian width is the FWHM of the optical-intensity envelope. Run the built-in validation suite with

```
python3 bragg_transfer_function_propagation_ui.py --self-test
```

3 Geometry and delayed two-photon coupling

Let $I(t)$ and $\phi(t)$ denote the source-side intensity envelope and optical phase. Taking the forward field at the atoms as the time reference, the reflected field samples the source a round-trip time earlier,

$$\tau_d = \frac{2L}{c}. \quad (1)$$

After removing constant propagation phases and assuming identical spatial modes, the complex two-photon coupling is

$$\tilde{\Omega}(t) = \Omega_{\text{in}} \frac{\sqrt{I(t)I(t-\tau_d)}}{I_{\text{pk}}} \exp\{i[\phi(t) - \phi(t-\tau_d)]\}. \quad (2)$$

Its modulus gives the raw envelope

$$\Omega_{\text{raw}}(t) = \Omega_{\text{in}} \frac{\sqrt{I(t)I(t-\tau_d)}}{I_{\text{pk}}}, \quad (3)$$

while its argument gives the local counter-propagating phase. Envelope overlap modifies the pulse shape and therefore H_{AI} . Phase differencing produces the separate delay filter D_d derived in section 7. The two effects enter at different stages of the calculation.

Definition of L

The software uses $\tau_d = 2L/c$, where L is one way. Entering a path length that already includes the round trip counts the factor of two twice. Additional fibre, electronic or cavity delays are not included automatically.

4 Square and Gaussian overlap

4.1 Square pulse

For a unit-height square command of duration τ centred on t_c , the common interval has duration

$$\tau_{\text{ov}} = \tau - \tau_d, \quad \tau > \tau_d, \quad (4)$$

and is centred at $t_c + \tau_d/2$. The raw rate is Ω_{in} inside that interval, so the area ratio is

$$\frac{A_{\text{raw}}}{A_0} = 1 - \frac{\tau_d}{\tau}. \quad (5)$$

If $\tau_d \geq \tau$, there is no common support and the software stops with an error.

4.2 Gaussian pulse

The Gaussian input convention is

$$I(t) = I_{\text{pk}} \exp\left[-\frac{(t - t_c)^2}{2\sigma^2}\right], \quad \tau_{\text{FWHM}} = 2\sqrt{2 \ln 2} \sigma. \quad (6)$$

Substitution into [equation \(3\)](#) gives

$$\Omega_{\text{raw}}(t) = \Omega_{\text{in}} \exp\left[-\frac{\tau_d^2}{8\sigma^2}\right] \exp\left[-\frac{(t - t_c - \tau_d/2)^2}{2\sigma^2}\right]. \quad (7)$$

The overlap is therefore a Gaussian of the same width, shifted by $\tau_d/2$. Its peak attenuation is the first exponential in [equation \(7\)](#). The calculation truncates each input envelope at $\pm 6\sigma$. The common numerical support then has total duration $12\sigma - \tau_d$, and its small missing-tail correction is retained through the error function.

5 Model A: pulse-area restoration

For every interaction j , Model A defines

$$\alpha_j = \frac{A_j^{\text{target}}}{A_j^{\text{raw}}}, \quad \Omega_{\text{eff}j}(t) = \alpha_j \Omega_{\text{raw}j}(t), \quad A_j^{\text{target}} \in \{\pi/2, \pi, \pi/2\}. \quad (8)$$

It represents an experimental recalibration of the local pulse area after propagation has been included.

For a square pulse, [equation \(5\)](#) gives

$$\alpha_{\text{sq}} = \frac{\tau}{\tau - \tau_d}, \quad \Omega_{\text{eff,pk}} = \Omega_{\text{in}} \alpha_{\text{sq}}. \quad (9)$$

The compensated pulse is shorter and higher than the no-delay pulse, so its Fourier envelope changes.

For an untruncated Gaussian, the area attenuation is exactly the peak attenuation. Hence

$$\alpha_G = \exp\left[\frac{\tau_d^2}{8\sigma^2}\right], \quad \Omega_{\text{eff}G}(t) = \Omega_{\text{in}} \exp\left[-\frac{(t - t_c - \tau_d/2)^2}{2\sigma^2}\right]. \quad (10)$$

Thus Model A restores the original Gaussian exactly, except for the common time shift. In the implemented $\pm 6\sigma$ representation, the normalization also includes the ratio of the nominal and common-support error functions. Because the removed tails are extremely small in the present example, the numerical result is indistinguishable from [equation \(10\)](#) at the plotted precision.

6 Sensitivity function

Define the accumulated area of interaction j as

$$\Theta_j(t) = \int_{t_{j,s}}^t \Omega_{\text{eff}j}(t') dt'. \quad (11)$$

Because Model A restores ideal areas, the MIGA/Cheinet sign convention gives

$$s(t) = \begin{cases} \sin \Theta_1(t), & t \text{ in the first } \pi/2 \text{ pulse,} \\ +1, & \text{between pulses 1 and 2,} \\ \cos \Theta_2(t), & t \text{ in the } \pi \text{ pulse,} \\ -1, & \text{between pulses 2 and 3,} \\ -\cos \Theta_3(t), & t \text{ in the last } \pi/2 \text{ pulse,} \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

The program evaluates the equivalent resonant two-level phase-step response. The two constructions agree for the restored $\pi/2 - \pi - \pi/2$ areas.

7 Transfer functions

Using

$$\mathcal{F}[x](\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt, \quad \omega = 2\pi f, \quad (13)$$

the atomic response to local Bragg phase is

$$H_{\text{AI}}(f) = n \int_{-\infty}^{\infty} \dot{s}(t) e^{-i2\pi f t} dt = n i 2\pi f \int_{-\infty}^{\infty} s(t) e^{-i2\pi f t} dt. \quad (14)$$

The second form follows by integration by parts because $s(t)$ vanishes outside the sequence. The code uses this second, numerically stable form. The delay between the two source-phase samples gives

$$D_d(f) = 1 - e^{-i2\pi f \tau_d}, \quad |D_d| = 2|\sin(\pi f \tau_d)|. \quad (15)$$

The source-phase and source-frequency responses are

$$H_{\text{src}\phi}(f) = H_{\text{AI}}(f) D_d(f), \quad H_{\nu}(f) = \frac{H_{\text{src}\phi}(f)}{if}. \quad (16)$$

The factor 2π cancels when frequency noise is specified in hertz because $\dot{\phi} = 2\pi\delta\nu$.

Table 1. Inputs, outputs and units of the displayed responses.

Quantity	Input → output	Unit	Interpretation
H_{AI}	local Bragg phase → atom phase	dimensionless	Atomic sequence and pulse-shape filtering
D_d	source phase → local phase difference	dimensionless	Round-trip delay differencer
$H_{\text{src}\phi}$	source phase → atom phase	dimensionless	Product $H_{\text{AI}}D_d$
H_ν	source frequency in Hz → atom phase	rad/Hz	Product $H_{\text{src}\phi}/(if)$

For a one-sided input PSD $S_x(f)$, the output PSD and variance are

$$S_\Phi(f) = |H_x(f)|^2 S_x(f), \quad \sigma_\Phi^2 = \int_0^\infty S_\Phi(f) df. \quad (17)$$

An amplitude spectral density is multiplied by $|H_x|$, not by $|H_x|^2$.

8 Numerical implementation

The schedule builder converts entered widths to nominal peak Rabi rates. For square pulses, $\Omega_{\pi/2} = (\pi/2)/\tau_{\pi/2}$ and $\Omega_\pi = \pi/\tau_\pi$. For Gaussian pulses, the target area is divided by $\sigma\sqrt{2\pi}\text{erf}(6/\sqrt{2})$. The delayed overlap, raw area and Model-A scaling are then calculated separately for each interaction.

The transfer routine evaluates $\int s(t)e^{-i\omega t} dt$. It uses Gauss–Legendre quadrature inside the pulses and integrates the long constant plateaus analytically with

$$\int_0^d e^{iqt} dt = d e^{iqd/2} \text{sinc}\left(\frac{qd}{2}\right), \quad \text{sinc}(x) = \frac{\sin x}{x}. \quad (18)$$

NumPy uses $\text{sinc}(x) = \sin(\pi x)/(\pi x)$, so the implementation passes $qd/(2\pi)$. Frequencies are processed in chunks and the quadrature order grows with the product of maximum frequency and pulse support.

✓ Validation

The built-in self-test verifies the zero-delay limit, the square overlap formula, Model-A area restoration, the analytic delay factor, $H_{\text{src}\phi} = H_{\text{AI}}D_d$, and $H_\nu = H_{\text{src}\phi}/(if)$ away from DC.

9 Worked case: square and Gaussian pulses

9.1 Common inputs

The example uses $n = 1$, $L = 150$ m, $T = 250$ ms, a $25\ \mu\text{s}$ $\pi/2$ width and a $50\ \mu\text{s}$ π width. For square pulses, these are command durations and both nominal rates are $\Omega/(2\pi) = 10$ kHz. For Gaussian pulses, they are intensity FWHM values; the corresponding σ values and area-normalized peaks are different. The frequency range is 0.1 Hz–100 kHz. The common round-trip delay is

$$\tau_d = 1.000\,692\ \mu\text{s}. \quad (19)$$

Table 2. Model-A pulse properties for the 150 m example. “Support” denotes the numerical common support; for Gaussian pulses it extends over the retained $\pm 6\sigma$ tails and is not a FWHM.

Shape and pulse	Input width (μs)	σ (μs)	Common support (μs)	α	Effective peak $\Omega/(2\pi)$
Square $\pi/2$	25	—	23.999308	1.041697	10.4170 kHz
Square π	50	—	48.999308	1.020423	10.2042 kHz
Gaussian $\pi/2$	25 FWHM	10.616523	126.397578	1.001111	9.3944 kHz
Gaussian π	50 FWHM	21.233045	253.795848	1.000278	9.3944 kHz

The square overlap loses 4.00% of the $\pi/2$ duration and 2.00% of the π duration. Model A compensates by raising the peaks to 10.4170 kHz and 10.2042 kHz. For the Gaussian $\pi/2$ and π pulses, the raw peak ratios are 0.998890 and 0.999722. The corresponding scale factors, 1.001111 and 1.000278, cancel those attenuations to the shown precision. Both effective Gaussian peaks are therefore 9.3944 kHz, equal to the nominal area-normalized input peak within rounding.

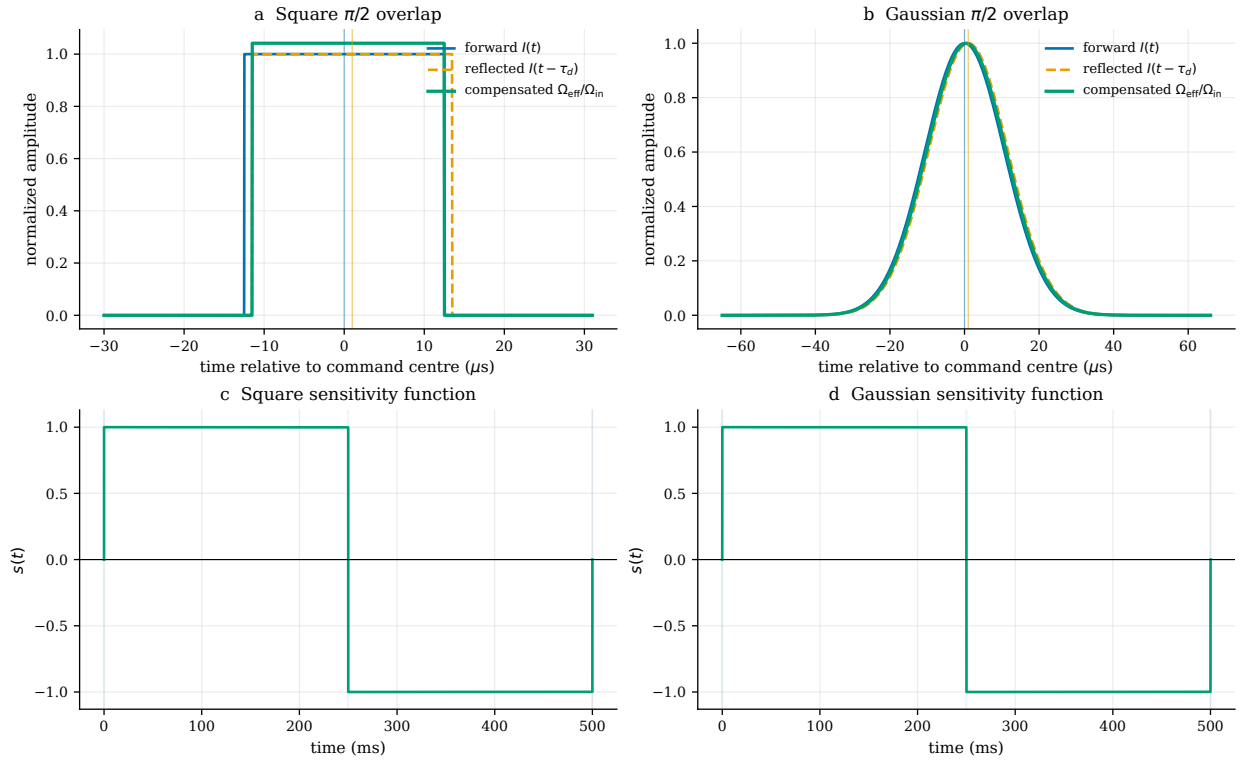


Figure 2. Model-A overlap and sensitivity functions. (a) The square envelopes have a visibly shortened common interval; area restoration raises the effective Rabi rate. (b) The delayed Gaussian overlap has the same width as the input Gaussian and is shifted by $\tau_d/2$; compensation restores its height. (c,d) On the full 500 ms sequence, both sensitivity functions have the same plateaus, while their pulse-edge shapes differ on a microsecond scale.

9.2 Atomic transfer function

At frequencies of order $1/T$, square and Gaussian responses are almost indistinguishable because the long free evolution dominates the microsecond pulse edges. At higher frequency, the envelope depends strongly on pulse shape. The square response develops sinc-like lobes and retains more high-frequency content. The smooth Gaussian response decays much more rapidly.

For the propagated square pulse, the shorter compensated duration shifts the high-frequency lobes relative

to the no-delay calculation. For the Gaussian pulse, [equation \(10\)](#) implies

$$H_{\text{AI,G}}^{(L)}(f) = e^{-i2\pi f\tau_d/2} H_{\text{AI,G}}^{(0)}(f) \quad (20)$$

in the untruncated limit. The propagation changes its complex phase but not its magnitude.

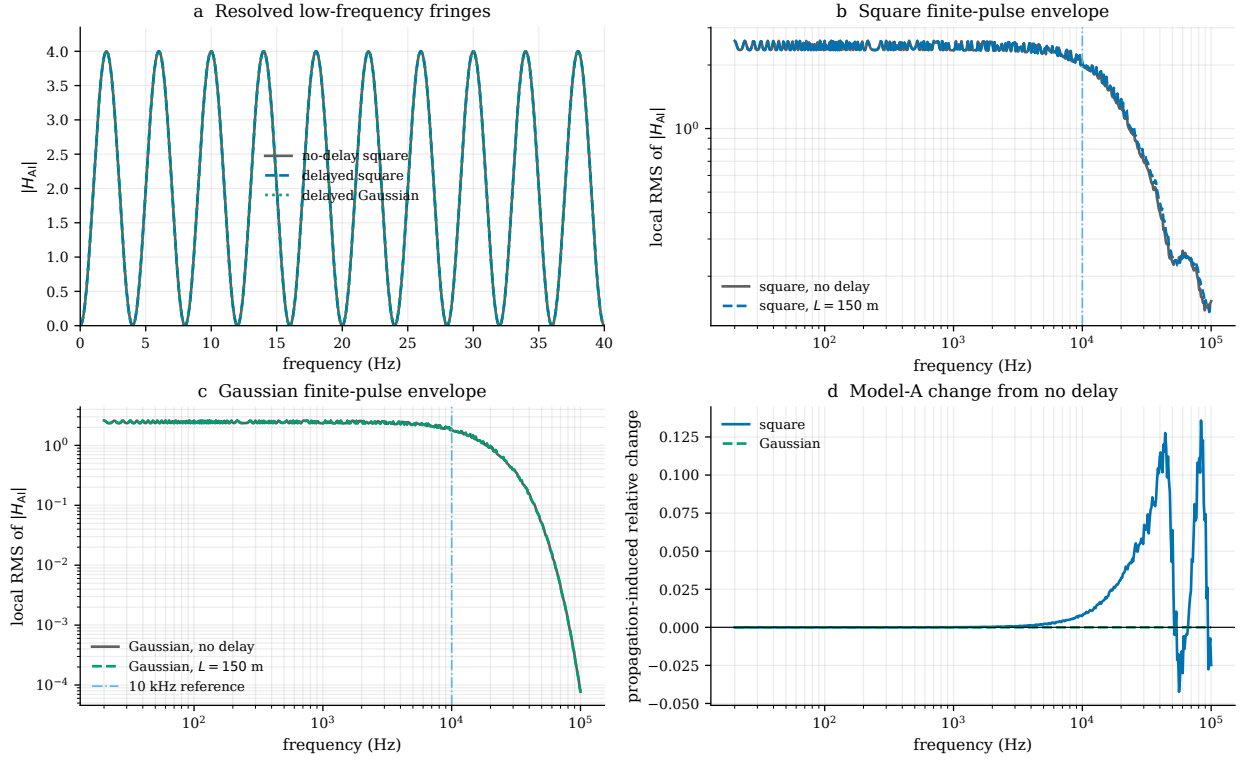


Figure 3. Atomic transfer functions under Model A. (a) The resolved low-frequency fringes coincide. (b) Square-pulse propagation modestly changes the finite-pulse envelope. (c) The delayed and no-delay Gaussian magnitudes overlap because area compensation restores the Gaussian shape. (d) The relative change confirms that the Gaussian magnitude is essentially invariant, whereas the square difference becomes appreciable near high-frequency lobes and zeros. Local RMS is used only to display the envelope without sensitivity to isolated fringe zeros.

Table 3. Selected ± 2 Hz local-RMS values of $|H_{\text{AI}}|$. The RMS window is a plotting diagnostic, not an additional physical filter.

Frequency	Square, $L = 0$	Square, $L = 150$ m	Gaussian, $L = 0$	Gaussian, $L = 150$ m
10 kHz	2.054095	2.070906	1.846185	1.846185
100 kHz	$1.452\,252 \times 10^{-1}$	$1.420\,195 \times 10^{-1}$	$7.707\,363 \times 10^{-5}$	$7.707\,417 \times 10^{-5}$

At 10 kHz, the square local RMS changes from 2.054095 to 2.070906, while the Gaussian value remains 1.846185 to six decimal places. At 100 kHz, the propagated square value is $1.420\,195 \times 10^{-1}$, whereas the Gaussian has already fallen to $7.707\,417 \times 10^{-5}$. This large difference is pulse-shape filtering, not an extra propagation penalty on the Gaussian.

9.3 Delay filter and total response

The phase-delay factor is common to both pulse shapes because L is the same. Its magnitude is 0.062865 at 10 kHz and 0.618448 at 100 kHz. The first non-DC zero occurs at $1/\tau_d \simeq 999.3$ kHz, outside the plotted range. Below it, $|D_d| \simeq 2\pi f\tau_d$.

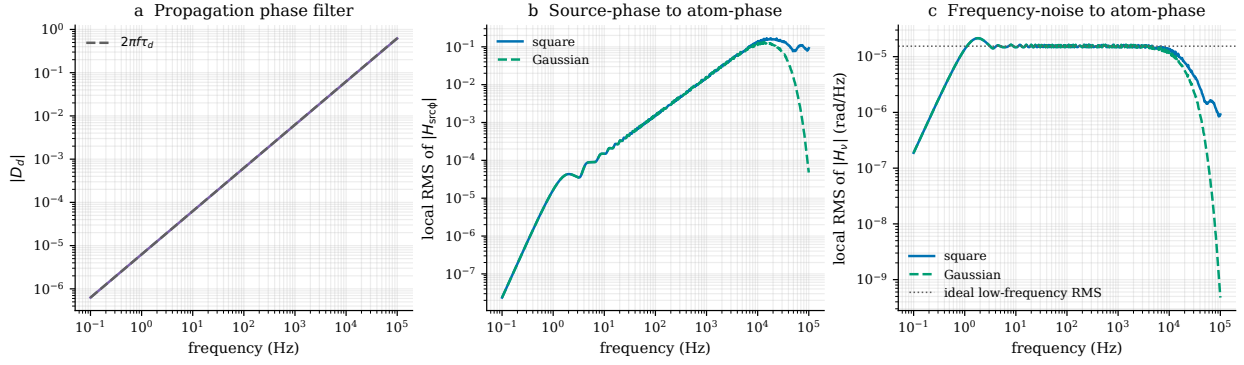


Figure 4. Combined propagation responses. (a) The delay differencer is identical for square and Gaussian pulses. (b) The source-phase response is $|H_{\text{src}\phi}| = |H_{\text{AI}}D_d|$; its high-frequency difference is inherited from the atomic pulse-shape filter. (c) The frequency-noise response is $|H_v| = |H_{\text{src}\phi}|/f$ in rad/Hz. The dotted level is the ideal-pulse low-frequency local-RMS limit $2\pi\tau_d\sqrt{6}$ for $n = 1$.

i Interpretation of the 150 m case

Under area compensation, propagation changes the square-pulse $|H_{\text{AI}}|$ because it shortens the interaction. It does not materially change the Gaussian $|H_{\text{AI}}|$ because the overlapped Gaussian has the same width and normalization restores its height. Nevertheless, both shapes acquire the same source-phase differencer D_d . Pulse-area calibration can restore the local Rabi envelope, but it cannot remove the delayed sampling of source laser phase.

10 Reproducing the figures

From this manual directory, run

```
python3 manual_figures_propagation.py
pdflatex -interaction=nonstopmode bragg_transfer_function_propagation_manual.tex
pdflatex -interaction=nonstopmode bragg_transfer_function_propagation_manual.tex
```

The plotting script imports the same numerical functions as the GUI, writes vector PDF and preview PNG figures, and regenerates `case_values.tex`. Upload the $\text{\LaTeX}$ source, `case_values.tex`, and the `figures` folder together when compiling on Overleaf.

11 Troubleshooting and limitations

Observation	Explanation and action
No square-pulse overlap	$\tau_d \geq \tau$ for at least one pulse. Reduce L or lengthen the pulse.
Gaussian support looks much longer than its entered width	The entered width is FWHM; the numerical support retains the tails out to $\pm 6\sigma$.
Gaussian propagation curves overlap exactly	This is the expected Model-A result in equation (10) , not a missing calculation. Inspect complex phase to observe the $\tau_d/2$ shift.
Square relative-change spikes	The no-delay response is near a lobe zero. Compare complex responses or a local envelope rather than a pointwise ratio.
H_{AI} is dimensionless but H_v is rad/Hz	They have different inputs: phase/phase versus phase/frequency.

Observation	Explanation and action
Required square peak is too high	The area-restoring factor grows as $\tau/(\tau - \tau_d)$. The assumed compensation may exceed available optical power.
Result differs from a multi-order simulation	The present backend is an effective resonant two-level model and excludes unwanted momentum orders and diffraction phases.

12 File map

File	Role
<code>bragg_transfer_function_propagation_ui.py</code>	GUI, delayed pulse construction, sensitivity function and transfer functions.
<code>PROPAGATION_MODEL.md</code>	Concise model definition and launch instructions.
<code>bragg_transfer_function_propagation_manual.tex</code> <code>manual_figures_propagation.py</code>	This Model-A square/Gaussian manual. Deterministic generator for the case figures and numerical macros.

References

- [1] P. Cheinet *et al.*, Measurement of the sensitivity function in a time-domain atomic interferometer, *IEEE Trans. Instrum. Meas.* **57**, 1141–1148 (2008). [doi:10.1109/TIM.2007.915148](https://doi.org/10.1109/TIM.2007.915148).
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- [3] B. Fang *et al.*, Improving the phase response of an atom interferometer by means of temporal pulse shaping, *New J. Phys.* **20**, 023020 (2018). [doi:10.1088/1367-2630/aaa37c](https://doi.org/10.1088/1367-2630/aaa37c).
- [4] J.-N. Siemß *et al.*, Analytic theory for Bragg atom interferometry based on the adiabatic theorem, *Phys. Rev. A* **102**, 033709 (2020). [doi:10.1103/PhysRevA.102.033709](https://doi.org/10.1103/PhysRevA.102.033709).
- [5] J. Jenewein *et al.*, Multiport atom interferometry with Bragg pulses, *Phys. Rev. A* **105**, 063316 (2022). [doi:10.1103/PhysRevA.105.063316](https://doi.org/10.1103/PhysRevA.105.063316).